

Application of artificial intelligence systems for stylometric analysis of texts as factor of sustainable development

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Abstract. Investment in human capital, along with natural resource management, is an important indicator of sustainable development. One of the areas of such investments is the creation of artificial intelligence systems that allow for the classification of texts. This paper analyzes the use of artificial intelligence systems for stylometric text analysis. On the basis of the algorithm of the convolutional artificial immune system, a system for stylometric analysis of texts was developed and implemented in software. In order to determine the possibility of using this system to determine the authorship of literary works, it was trained and tested. For this, the works of two authors were chosen: Leo Tolstoy and Fyodor Kryukov. This system demonstrated a high quality of text classification and a good speed of work and learning. So, to test the performance of the system, 11 works by Leo Tolstoy and 12 works by Fedor Kryukov were taken that were not used to train the system. All works of these authors were classified correctly. It should be noted that the artificial immune system algorithm can also be successfully used in other tasks requiring text classification. **Keywords:** sustainable development, artificial intelligence, machine learning, pattern recognition, text processing.

1 Introduction

Sustainable development requires not only the rational use of natural, primarily non-renewable resources, but also an increase in investment in human capital. One of the areas of such investments is the creation of artificial intelligence systems that allow for stylometric analysis of texts.

Stylometric analysis of texts is an important tool for literary, historical and forensic research. The use of this method allows both the attribution of texts and the determination of the time of their writing. One of the directions of stylometric analysis of texts is the identification of the author's invariant, a quantitative characteristic that will be similar for the texts of one author and different for texts of different authors. Before the advent of artificial intelligence systems, the most common stylometry methods were statistical methods such as:

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The average number of words in a sentence, the percentage of function words (particles, prepositions, conjunctions), the average number of syllables in a word, the frequency of use of nouns, adjectives, verbs, etc.

Currently, artificial intelligence systems, and in particular artificial neural networks, are increasingly used to conduct stylometric analysis. It should be noted that in practice the definition of authorship is only one of the areas of stylometric research. Let's consider these directions.

2 Materials and Methods

The first direction is to determine the politeness of the text. For example, a statement like: "Could you please reply to this letter?" is more polite than "Answer this email!". Thus, the authors of [1] obtained a data set of 1.39 million text instances automatically labeled according to the degree of politeness. However, it should be borne in mind that politeness depends on culture and different countries have their own criteria for politeness.

Another direction is the definition of formality and non-formality of texts. For the first time, the degree of formality of texts was studied in [2]. The paper [3] presents the Yahoo Answers Formality Corpus (GYAFC) formality dataset containing 50,000 pairs of formal and informal sentences. These data were obtained in the following way. First, 50 thousand informal sentences were taken from the Yahoo Answers corpus in English, then a formal sentence was written for each of them by people. In [4], a data set with formal and informal sentences is presented for three languages: French, Brazilian Portuguese, and Italian.

In [5], a corpus consisting of pairs of sentences is implemented: biased and corresponding neutral ones. This corpus is assembled from Wikipedia edits that adjusted the tone of biased sentences to neutral sentences.

Features of texts related to the gender of the authors of these texts are studied in detail in numerous works [6-10]. A set of 2.5 million texts marked by gender, designed for training neural networks, was proposed in [11].

There are also works that explore the issue of classifying texts according to their topic. So in [12], using Yahoo! Answers, a data set consisting of two types of texts has been formed: entertainment and politics. In [13], a set is proposed that includes texts belonging to 21 classes, including: sports, politics, science and practice.

The classification of reviews according to their tonality is studied in detail in [14] (Yelp reviews are investigated) and [15] (Amazon product reviews are investigated). As a rule, all reviews are ranked according to the change in the polarity of the mood in the reviews, for example, from positive reviews to negative.

The paper [16] studies the use of an artificial neural network to identify offensive texts. The authors of this work collected a data set consisting of 350 thousand offensive sentences and 7 million non-offensive sentences.

The task of classifying texts according to their political orientation is investigated in [17]. In particular, the dependence of the text style on whether its author belongs to the Republican or Democratic Party in the USA is investigated.

Another application of artificial neural networks is the transformation of complex texts into texts written in a more understandable language. Such a task may arise if it is necessary to adapt narrow professional texts, such as medical or legal, for a wider non-professional audience [18]. The work [19] considers the transformation of standard Wikipedia texts into a simpler version of Wikipedia, the so-called Simple Wikipedia. Another frequently researched task is to simplify medical texts to the point where they can be understood by patients. This problem was studied in detail in [20 - 22].

The issue of determining the authorship of texts is also studied in sufficient detail [23]. The work [24] presents 18 thousand pairs consisting of sentences from the original works of William Shakespeare and their corresponding sentences translated into modern English.

3 Results

As can be seen from the previous chapter, currently artificial neural networks are usually used for stylometric analysis of texts. The popularity of this algorithm is mainly due to two factors.

Firstly, artificial neural networks provide, perhaps, the best quality of text recognition of all known machine learning methods.

Secondly, and no less important, there are quite a large number of libraries in which these methods are implemented. Accordingly, neural networks can be created and used by a wide range of users who are not too versed in machine learning algorithms.

With all the advantages, however, artificial immune systems have one very significant drawback – a long training time.

This paper presents a system for stylometric analysis of texts based on a new machine learning algorithm - a convolutional artificial immune system. The development of algorithms for artificial immune systems is a promising area of artificial intelligence [25–26]. Compared to the most commonly used artificial neural network algorithms at present, the convolutional artificial immune system algorithm has a much faster learning rate.

In order to determine the possibility of using this algorithm to determine the authorship of literary works, the system was trained. For this, the works of two authors were chosen: Leo Tolstoy and Fyodor Kryukov.

The works of Leo Tolstoy taken for training the convolution of the artificial immune system: *Anna Karenina*; *Childhood*; *Devil*; *Cossacks*.

The works of Fedor Kryukov taken for training: *Weekdays*; *Two worlds*; *The soul is one*; *Thirst*; *Swell*; *Italian Zamchalov*; *To the source of healing*; *Cossack (From the village life)*; *Cossack stanitsa courts*; *Mother*; *A glimpse*.

System testing was conducted on an Acer Aspire 5 laptop (AMD Ryzen 7 4700U processor, 12 GB RAM). The system training time was 103.009 s.

To determine the class to which the text belongs, the ratio was calculated s_k :

$$s_k = \frac{q_k^1}{q_k^0} \quad (1)$$

here, q_k^0 - is the initial number of antibodies of the corresponding class, k - is the number of the corresponding class, q_k^1 - is the number of antibodies in these classes after interaction with the classified text.

It was believed that the text belongs to the class for which the value s_k has the greatest value.

To test the system's operability, 11 works by Leo Tolstoy and 12 works by Fyodor Kryukov were taken, which were not used for training the system. The results of the system operation are presented in Table 1.

Table 1. Analysis of the texts of the works Leo Tolstoy and Fedor Kryukov.

№	Title of the work and author	Value of s_k Fedor Kryukov	Value of s_k Leo Tolstoy	Classification time, s
1	Hadji Murad (Leo Tolstoy)	0.401	1.588	2.173
2	Shark (Leo Tolstoy)	0.896	1.102	1.862
3	After the Ball (Leo Tolstoy)	0.341	1.646	1.862
4	Family Happiness (Leo Tolstoy)	0.179	1.851	2.144
5	Sevastopol stories (Leo Tolstoy)	0.852	1.152	2.237
6	Death of Ivan Ilyich (Leo Tolstoy)	0.342	1.637	2.022
7	Sunday (Leo Tolstoy)	0.22	1.812	3.256
8	Youth (Leo Tolstoy)	0.299	1.724	2.320
9	Notes of a Madman (Leo Tolstoy)	0.084	1.913	1.936
10	Living corpse (Leo Tolstoy)	0.161	1.811	1.959
11	War and Peace (Leo Tolstoy)	0.165	1.83	4.209
12	Without fire (Fedor Kryukov)	1.733	0.26	1.922
13	In the depths (Fedor Kryukov)	1.616	0.38	2.161
14	In cell no. 380 (Fedor Kryukov)	1.233	0.77	1.886
15	In native places (Fedor Kryukov)	1.301	0.702	1.945
16	In the snowdrifts (Fedor Kryukov)	1.946	0.058	2.278
17	In the corner (FedorKryukov)	1.434	0.553	2.671
18	Meeting (FedorKryukov)	1.253	0.743	2.513
19	Group B (FedorKryukov)	1.032	0.969	2.794
20	Gulebshchiki (FedorKryukov)	1.447	0.55	2.472
21	On the quiet Don (Fedor Krjukov)	1.568	0.429	2.186
22	Squall (Fedor Krjukov)	1.153	0.852	2.109
23	Shulgin massacre (FedorKryukov)	1.329	0.663	2.466

The first column of this table shows the number of the experimental study. The second column of this table contains the names of literary and works and their authors. In the third column is the value of the s_k ratio for the assumption that the author of the corresponding work is Fyodor Kryukov. In the fourth column is the value of the s_k ratio for the assumption that the author of the corresponding work is Leo Tolstoy. The fifth column shows the time for which the system classified the text.

4 Conclusions

As can be seen from this table, the quality of text classification by the system is high. Not a single mistake was made in determining the authorship of the works.

The average text classification time in this experiment is 2.321 s. At the same time, the average classification time of a work written by Leo Tolstoy was 2.362 seconds. The average classification time of a work written by Fyodor Kryukov is 2.284s. The difference in the time of classification of the works of these authors is due to the fact that on average the volume of works written by Leo Tolstoy is greater than the average volume of works written by Fyodor Kryukov.

It should be noted that the algorithm of the artificial immune system can also be successfully used in other tasks requiring text classification. For example, when processing medical information or identifying illegal content on social networks.

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